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Ke Qiu

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EDUCATION

PhD Candidate | Zhejiang University *September 2022 – July 2027 (expected)*
Control Science and Engineering (Robotics).
Supervisor: Profs. Yue Wang, Haojian Lu, Rong Xiong

BEng (Chu Kochen Honors Program) | Zhejiang University *September 2018 – July 2022*
Mechanical Engineering. Minor in Mathematics.
GPA 3.96/4.0 (89.84/100), Top 5%

RESEARCH EXPERIENCE

External force estimation in continuum robots: Closed-form solution, solver and validation [Video]

- Derived a closed-form solution for external force estimation of Kirchhoff rods, and extended to multisectional tendon-driven continuum robots. Developed a high-order state estimation algorithm with Gaussian process regression, where the prior mean and kernel functions are derived from stochastic differential equations, and discrete noisy pose/strain measurements are fused into a posterior state distribution. Gauss–Newton algorithm for sparse block tridiagonal linear systems reduces time complexity to O(n). Our algorithm achieves superior estimation accuracy and noise robustness, and a 10x speedup in computational efficiency, compared to conventional non-linear solvers (e.g., BFGS). Real-time applications are demonstrated.

An actuator space optimal kinematic path tracking framework for tendon-driven continuum robots: Theory, algorithm and validation [Paper] [Code]

- Based on the inverse kinematics solver, formulated linear actuator mapping and applied dynamic programming to find the shortest actuator path. Derived optimal time allocation considering velocity and acceleration constraints. Designed and implemented a control framework, where trajectories planned offline serve as feedforward and inverse Jacobian computed online as real-time feedback, achieving accurate tip pose tracking. Developed an asynchronous system using C++/UDP and PLC/EtherCAT for >100 Hz stable real-time closed-loop control.

An efficient multi-solution solver for the inverse kinematics of 3-section constant-curvature robots [Paper] [Page]

- Formulated an analytical reduction theory that compresses a highly non-linear system of equations into a one-dimensional problem. Implemented a bounded error approximation to traverse the feasible space with guaranteed resolution completeness. Developed a solver in Matlab that eliminates initial values and local minima dependency of traditional non-linear optimisers, achieving 85% reduction in computational latency while providing multiple solutions.

Riemannian optimisation for kinematic closed-chain robots

- Modelled kinematic closed-chain robot systems as Riemannian submanifolds, constructing a retraction to perform state updates under high-dimensional non-linear equality constraints. Resolved strict inequality boundaries (e.g., joint limits) by projecting gradients onto tangent cones. Our algorithm improves convergence speed and numerical stability for motion planning of highly redundant closed-chain mechanisms.

INTERNSHIP EXPERIENCE

Research Intern | Zhejiang Humanoid Robot Innovation Center Co. Ltd., Ningbo, China

- Dynamic parameter identification and control optimisation *November 2024 – January 2025*

Developed a regression algorithm to identify latent system parameters, including joint friction and damping coefficient. Designed robust data collection and statistical fitting pipelines. Analyse the frequency spectra of reinforcement learning control signals via FFT, according to which the control bandwidths are optimised and the optimal ranges of control parameters are tested experimentally.

- Inverse kinematics analysis of parallel joint mechanisms *July 2024 – August 2024*

Modelled non-linear constrained kinematics for parallel-link ankle and waist joint mechanisms. Developed a numerical solving algorithm and implemented a C++ toolkit that achieves stable computation at 500 Hz. Analysed statics of both serial and parallel mechanisms, solving for the boundaries of permissible workspace and analysing torque transmission for motion planning and motor selection.

Research Intern | ABB Corporate Research Center, ABB Engineering (Shanghai) Ltd., Shanghai, China

- High-precision on-site calibration for industrial manipulators *July 2021 – August 2021*

Applied Lie group $SE(3)$ and Lie algebra $se(3)$ to build geometric models for 6DOF industrial manipulators. Constructed a non-linear least-squares problem using on-site canonical blocks, and calibrated latent model parameters by using the Levenberg–Marquardt algorithm, which reduces absolute positioning errors.

PUBLICATIONS

External force estimation in continuum robots: Closed-form solution, solver and validation
Under Review. Submitted to *The International Journal of Robotics Research*, 2026
- Ke Qiu, Fei Wang, Sven Lilge, Yujie Cui, Rong Xiong, Haojian Lu, Yue Wang

An actuator space optimal kinematic path tracking framework for tendon-driven continuum robots: Theory, algorithm and validation
The International Journal of Robotics Research, 2025
- Ke Qiu, Hongye Zhang, Jingyu Zhang, Rong Xiong, Haojian Lu, Yue Wang

An efficient multi-solution solver for the inverse kinematics of 3-section constant-curvature robots
Robotics: Science and Systems, 2023
- Ke Qiu, Jingyu Zhang, Danying Sun, Rong Xiong, Haojian Lu, Yue Wang

OTHERS (* indicates equal contribution)

Learning-based dynamics modeling and robust control for tendon-driven continuum robots
International Conference on Intelligent Robots and Systems (IROS), 2026
- Ziqing Zou*, Ke Qiu*, Fei Wang, Haojian Lu, Rong Xiong, Yue Wang

Learning the inverse kinematics of magnetic continuum robot for teleoperated navigation
International Conference on Intelligent Robots and Systems (IROS), 2024
- Pingyu Xiang*, Ke Qiu*, Danying Sun, …, Yue Wang, Rong Xiong, Haojian Lu

Reinforcement learning of serpentine locomotion for a snake robot
International Conference on Real-time Computing and Robotics (RCAR), 2021
- Ke Qiu, Hang Zhang, Yikai Lv, Yunkai Wang, Chunlin Zhou, Rong Xiong

AWARDS

IROS 2025 Best Paper Award Finalists	*2025*
A4X Robotics Scholarship	*2025*
Outstanding Postgraduate Student, Zhejiang University	*2023, 2025*
Outstanding Graduate of Zhejiang Province, Zhejiang Provincial Government	*2022*
National Scholarship, Ministry of Education, China	*2021*
Innovation Scholarship, Chu Kochen Honors College	*2021, 2020*
First-Class Scholarship, Zhejiang University	*2020, 2019*
First-Class Scholarship, Department of Mechanical Engineering, Zhejiang University	*2020, 2019*

PROFESSIONAL EXPERIENCE

Reviewer for
- IEEE Transactions on Robotics (T-RO)
- IEEE/ASME Transactions on Mechatronics (TMECH)
- IEEE Transactions on Automation Sciences and Engineering (T-ASE)
- Cyborg and Bionic Systems (Science Partner Journal)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

REFERENCE

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Department of Control Science and Engineering

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Prof. Rong Xiong
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教育经历

博士(在读) | 浙江大学 *2022 年 9 月 – 2027 年 7 月 (预计)*
控制科学与工程(机器人方向). 导师: 王越，陆豪健，熊蓉

学士(竺可桢荣誉项目) | 浙江大学 *2018 年 9 月 – 2022 年 7 月*
机械工程. 辅修数学. 均绩 3.96/4.0，均分 89.84/100，排名前 5%

研究经历

External force estimation in continuum robots: Closed-form solution, solver and validation [Video]
- 提出 Kirchhoff 杆的外力估计闭式解理论，并推广到多段线驱动连续体机器人. 设计基于高斯过程回归的高阶状态估计算法，从随机微分方程导出先验均值与核函数，融合离散测量获得状态后验分布. 回归采用 Gauss–Newton 法求解稀疏分块三对角线性系统，时间复杂度 O(n). 对比实验表明该算法具有更高力估计精度和噪声鲁棒性，计算效率较传统数值优化(如 BFGS)提升约一个数量级. 有效应用于动态力感知与临床场景.

An actuator space optimal kinematic path tracking framework for tendon-driven continuum robots: Theory, algorithm and validation [Paper] [Code]
- 基于逆运动学求解器，推导线性的驱动映射，使用动态规划求解驱动器空间最短距离路径，考虑速度加速度约束分配最短时间. 搭建前馈反馈控制框架，以离线最优轨迹为前馈，在线计算逆雅可比反馈，实现末端位姿的高精度跟踪. 搭建上位机(C++/UDP)和 PLC(ST/EtherCAT)高实时异步系统，实现>100 Hz 稳定闭环控制.

An efficient multi-solution solver for the inverse kinematics of 3-section constant-curvature robots [Paper] [Page]
- 针对三段等曲率建模的连续体机器人非线性逆运动学方程，推导解析条件将原方程等价简化为一维问题，引入有界误差近似从而遍历求解，保障分辨率完备性. 开发 Matlab 求解器，相比传统数值求解器无需提供初值，改善成功率，总体计算效率提升 85%，同时支持多解输出.

Riemannian optimisation for kinematic closed-chain robots
- 将运动学闭环机器人建模为黎曼子流形，构造拉回映射实现高维非线性等式约束下的状态更新，针对不等式约束(如关节限位)，将问题转化为切锥上的投影梯度下降，提升高冗余闭环机构运动规划速度与稳定性.

实习经历

研究实习生 | 浙江人形机器人创新中心有限公司，宁波 (累计融资 22 亿元，人形机器人头部企业)
- 动力学参数辨识与控制参数优化 *2024 年 11 月 – 2025 年 1 月*

实现关节摩擦力与阻尼系数的参数辨识算法，设计实验进行数据采集与拟合计算. 对强化学习策略输出的控制信号进行频域分析(FFT)，据此优化底层驱动器带宽，通过实验确定最佳运动控制参数范围.

- 并联关节机构的逆运动学分析 *2024 年 7 月 – 2024 年 8 月*

建模踝关节与腰关节并联连杆机构的非线性带约束运动学，推导数值求解算法，开发 C++工具包实现 500 Hz 稳定解算. 分析串并联机构静力学，求解容许工作空间边界及力矩耦合关系，优化运动规划与电机选型.

研究实习生 | ABB 中国研究院, ABB 工程(上海)有限公司，上海 (世界 500 强企业)
- 工业机械臂现场高精度标定算法 *2021 年 7 月 – 2021 年 8 月*

基于李群与李代数建模 6DOF 工业机械臂运动学，使用现场标准工件提供的末端位置约束构造非线性最小二乘问题，使用 LM 算法迭代求解各关节真实转轴参数，实现现场快速标定，补偿模型误差提升绝对定位精度.

发表论文

External force estimation in continuum robots: Closed-form solution, solver and validation
Under Review. Submitted to *The International Journal of Robotics Research*, 2026
- Ke Qiu, Fei Wang, Sven Lilge, Yujie Cui, Rong Xiong, Haojian Lu, Yue Wang

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其它（* 代表共同第一作者）

Learning-based dynamics modeling and robust control for tendon-driven continuum robots
International Conference on Intelligent Robots and Systems (IROS), 2026
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Learning the inverse kinematics of magnetic continuum robot for teleoperated navigation
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- Pingyu Xiang*, Ke Qiu*, Danying Sun, …, Yue Wang, Rong Xiong, Haojian Lu

Reinforcement learning of serpentine locomotion for a snake robot
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- Ke Qiu, Hang Zhang, Yikai Lv, Yunkai Wang, Chunlin Zhou, Rong Xiong

荣誉

IROS 2025 最佳论文提名	*2025*
积加机器人奖学金	*2025*
优秀研究生，浙江大学	*2023, 2025*
浙江省优秀毕业生，省政府	*2022*
国家奖学金，教育部	*2021*
竺可桢学院创新奖学金，竺可桢学院	*2021, 2020*
浙江大学一等奖学金，浙江大学	*2020, 2019*
国际精密一等奖学金，机械工程学院	*2020, 2019*

学术兼职

作为以下期刊及会议审稿人
- IEEE Transactions on Robotics (T-RO)
- IEEE/ASME Transactions on Mechatronics (TMECH)
- IEEE Transactions on Automation Sciences and Engineering (T-ASE)
- Cyborg and Bionic Systems (Science Partner Journal)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

参考信息

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